



**NICHOMACHINES.COM**



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# Startup procedure

When start up on the machine:

1. Switch ON main power.
2. When the PC Screen is ON
3. Press Servo Start
4. Wait for the light, in the light tower to light UP.
5. When the machine towerlight is ON



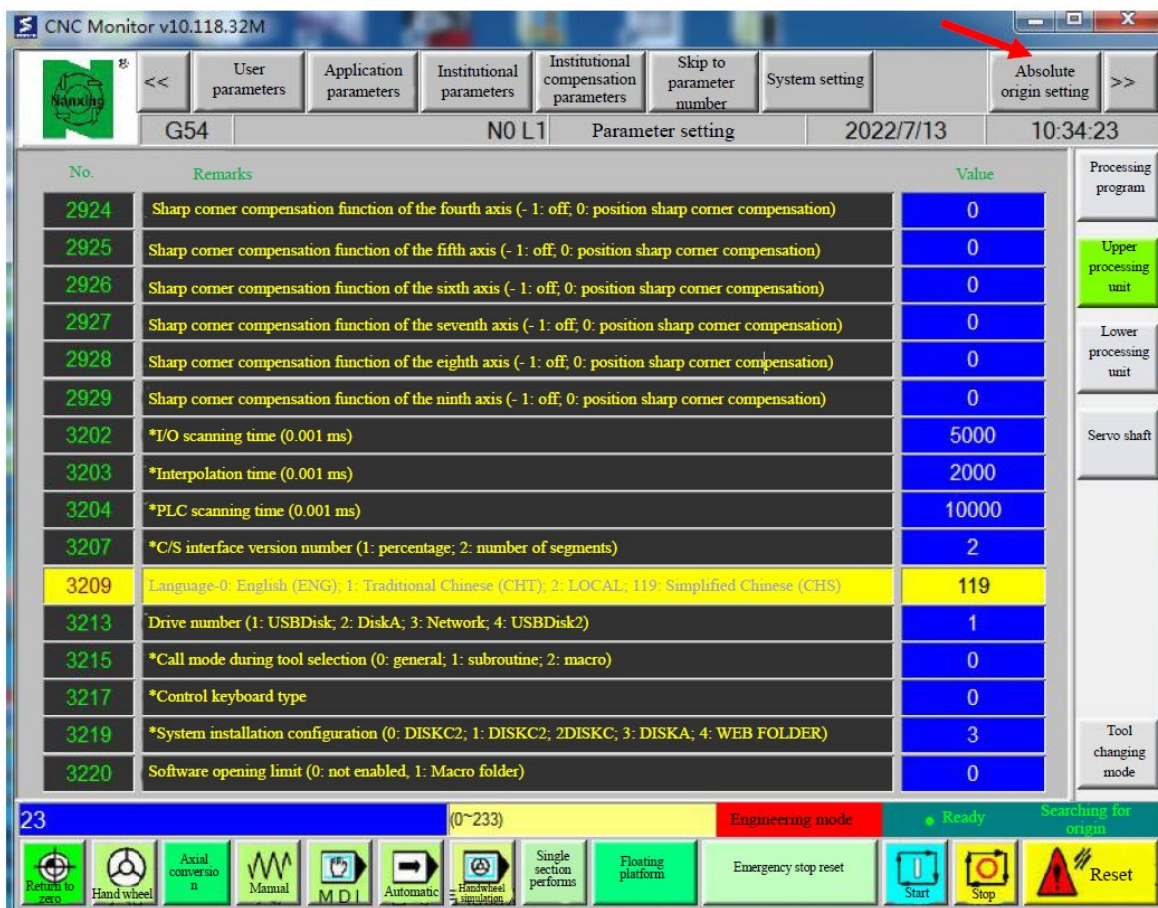
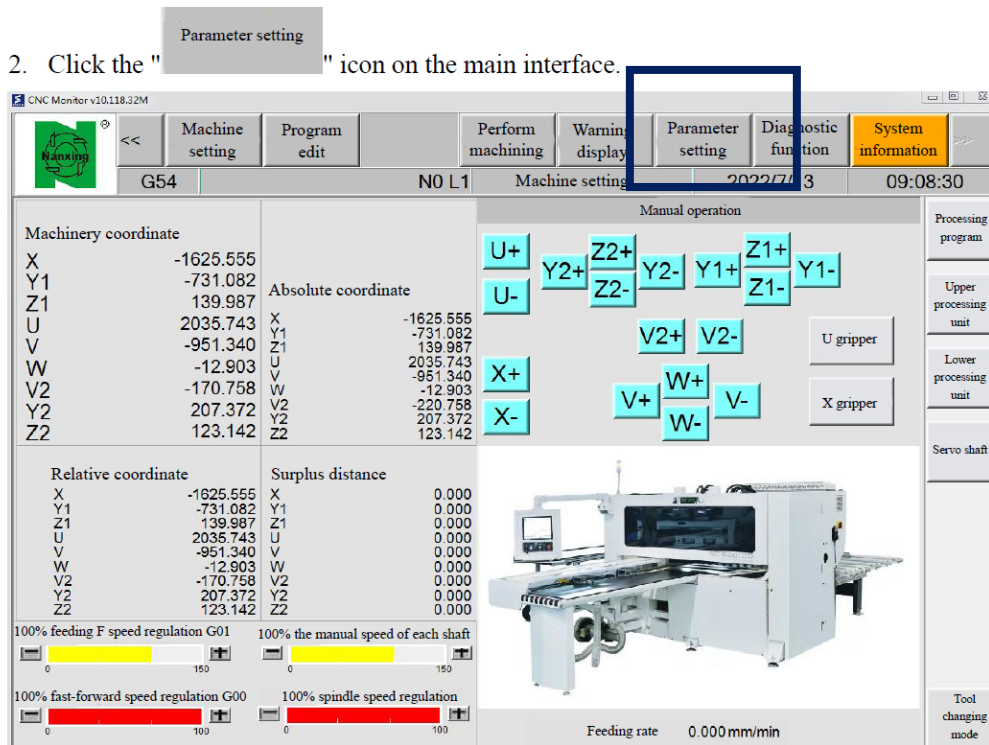
6. Start up cnc.bat
7. Open up the software manual for setting up the machine.



## Laugauge change

For changing of the language go to: Parameter setting, find parameter 3209 and press 0 for English, the password if needed is: 520

- Click the "Parameter setting" icon on the main interface.





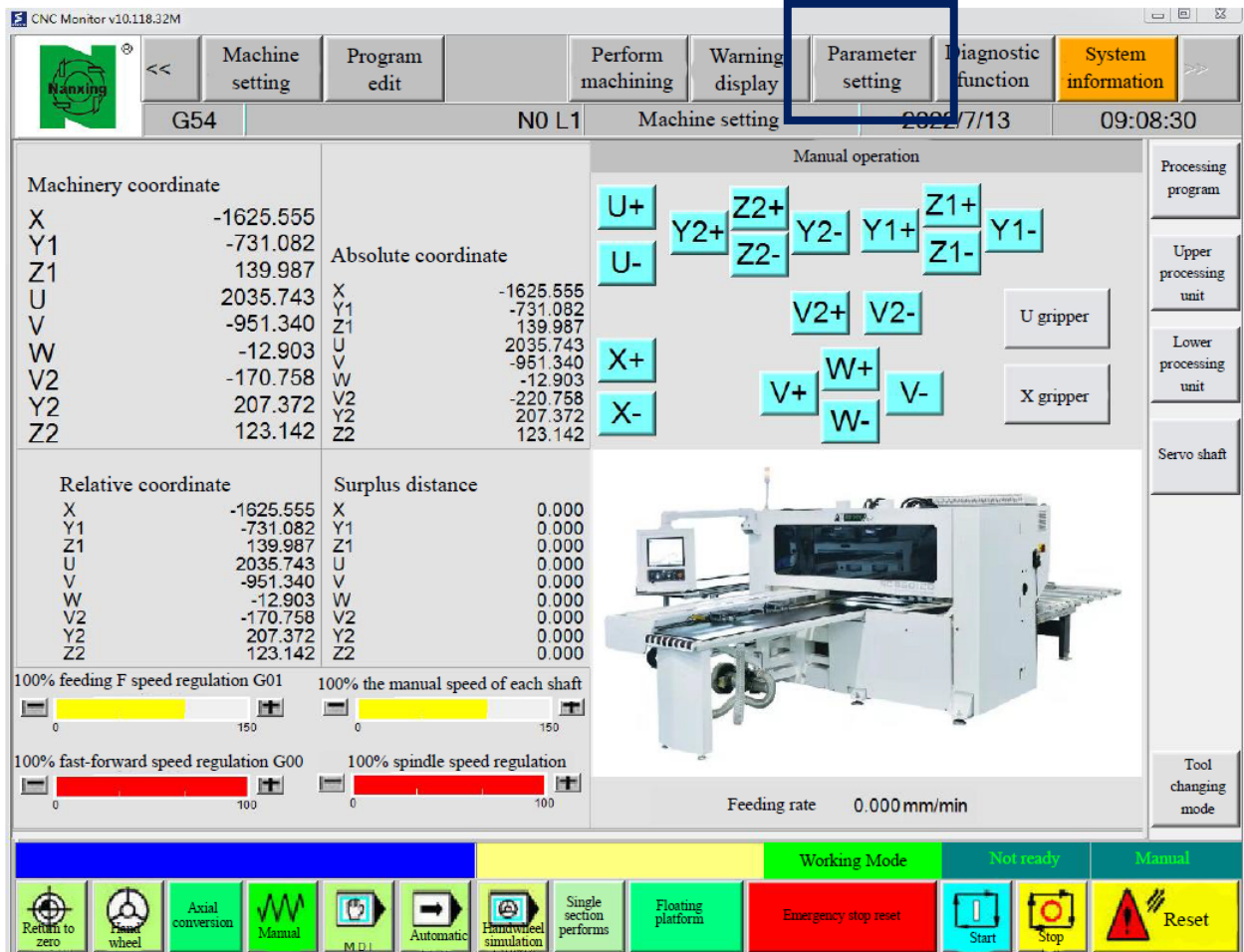
## Origin position setting

For absolute coordinates, the steps for setting the origin of each axis are as follows:

Manually move the axis to the target zero (origin) position.

Parameter setting

2. Click the " " icon on the main interface.



Go to parameter 3418 (Normal mode and Engineering mode) and change it to 1.

Code again here is 520



Enter absolute position setting.

CNC Monitor v10.118.32M

Navigation: << User parameters Application parameters Institutional parameters Institutional compensation parameters Skip to parameter number System setting **Absolute origin setting** >>

Current: G54 N0 L1 Parameter setting 2022/7/13 10:34:23

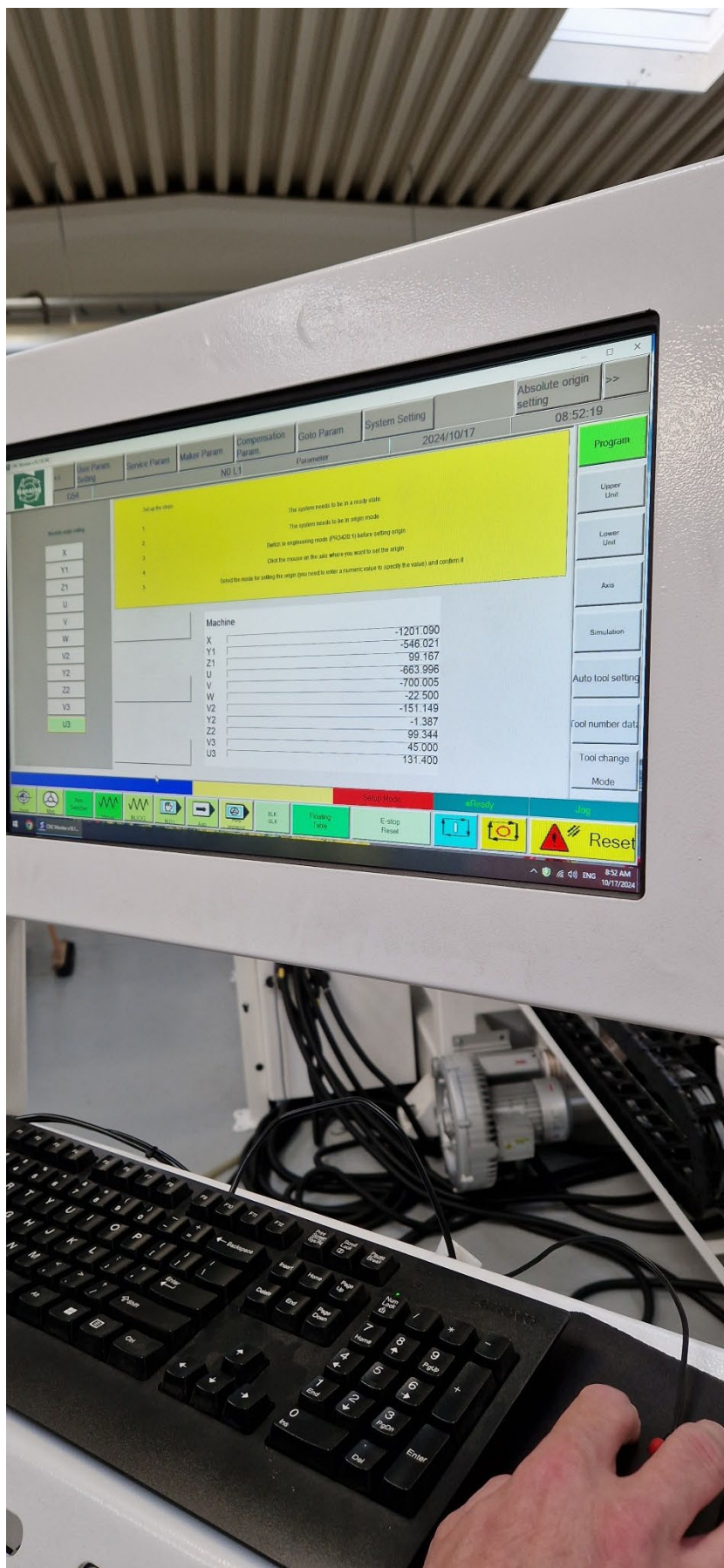
| No.  | Remarks                                                                                                  | Value |
|------|----------------------------------------------------------------------------------------------------------|-------|
| 2924 | Sharp corner compensation function of the fourth axis (- 1: off, 0: position sharp corner compensation)  | 0     |
| 2925 | Sharp corner compensation function of the fifth axis (- 1: off, 0: position sharp corner compensation)   | 0     |
| 2926 | Sharp corner compensation function of the sixth axis (- 1: off, 0: position sharp corner compensation)   | 0     |
| 2927 | Sharp corner compensation function of the seventh axis (- 1: off, 0: position sharp corner compensation) | 0     |
| 2928 | Sharp corner compensation function of the eighth axis (- 1: off, 0: position sharp corner compensation)  | 0     |
| 2929 | Sharp corner compensation function of the ninth axis (- 1: off, 0: position sharp corner compensation)   | 0     |
| 3202 | *I/O scanning time (0.001 ms)                                                                            | 5000  |
| 3203 | *Interpolation time (0.001 ms)                                                                           | 2000  |
| 3204 | *PLC scanning time (0.001 ms)                                                                            | 10000 |
| 3207 | *C/S interface version number (1: percentage; 2: number of segments)                                     | 2     |
| 3209 | Language-0: English (ENG); 1: Traditional Chinese (CHT); 2: LOCAL; 119: Simplified Chinese (CHS)         | 119   |
| 3213 | Drive number (1: USBDisk; 2: DiskA; 3: Network; 4: USBDisk2)                                             | 1     |
| 3215 | *Call mode during tool selection (0: general; 1: subroutine; 2: macro)                                   | 0     |
| 3217 | *Control keyboard type                                                                                   | 0     |
| 3219 | *System installation configuration (0: DISK2; 1: DISK2; 2DISK; 3: DISKA; 4: WEB FOLDER)                  | 3     |
| 3220 | Software opening limit (0: not enabled, 1: Macro folder)                                                 | 0     |

Processing program: Upper processing unit, Lower processing unit, Servo shaft, Tool changing mode

23 (0~233) Engineering mode Ready Searching for origin

Buttons: Return to zero, Hand wheel, Axial conversion, Manual, MDI, Automatic, Handwheel simulation, Single section performs, Floating platform, Emergency stop reset, Start, Stop, Reset



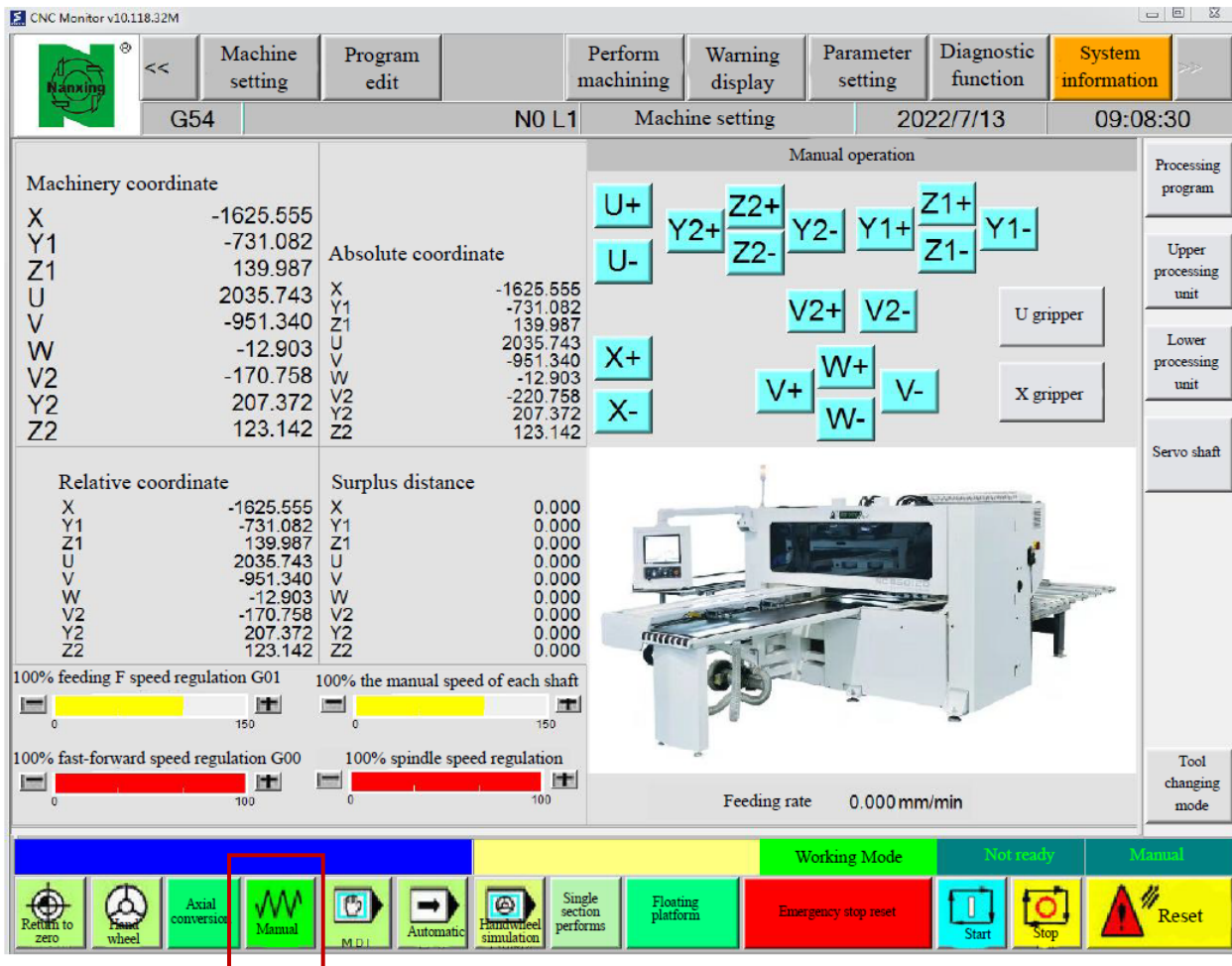


## Origin setting – Clamps / grippers.

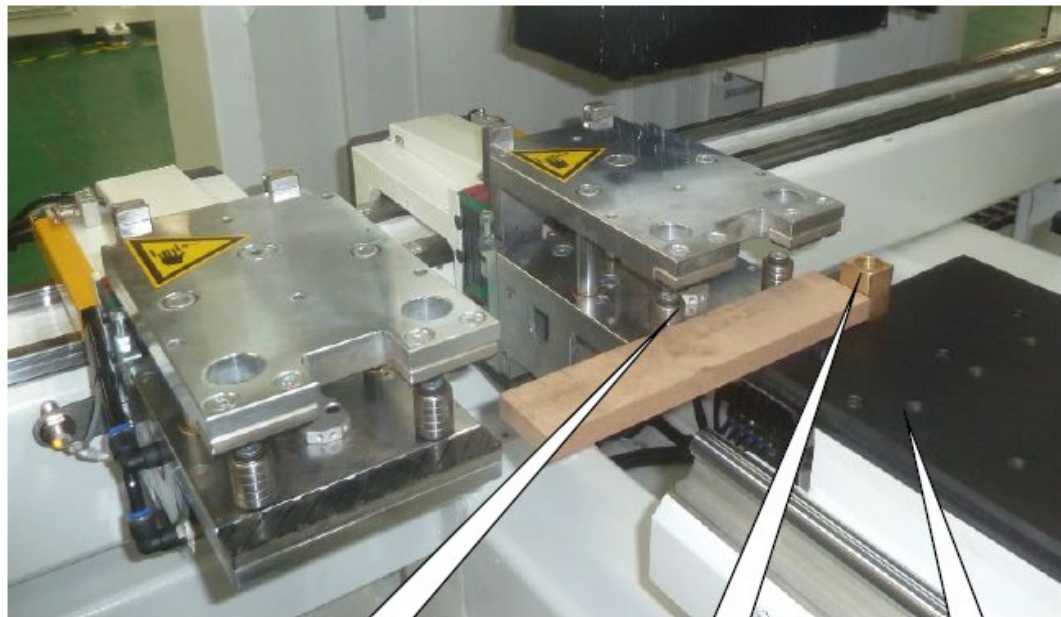
For changing the origin position of the clamps, do the following steps:

1. Move the clamps manually by entering the manual mode, move the X clamp first and then U clamp afterwards.

2. Click the "Parameter setting" icon on the main interface.

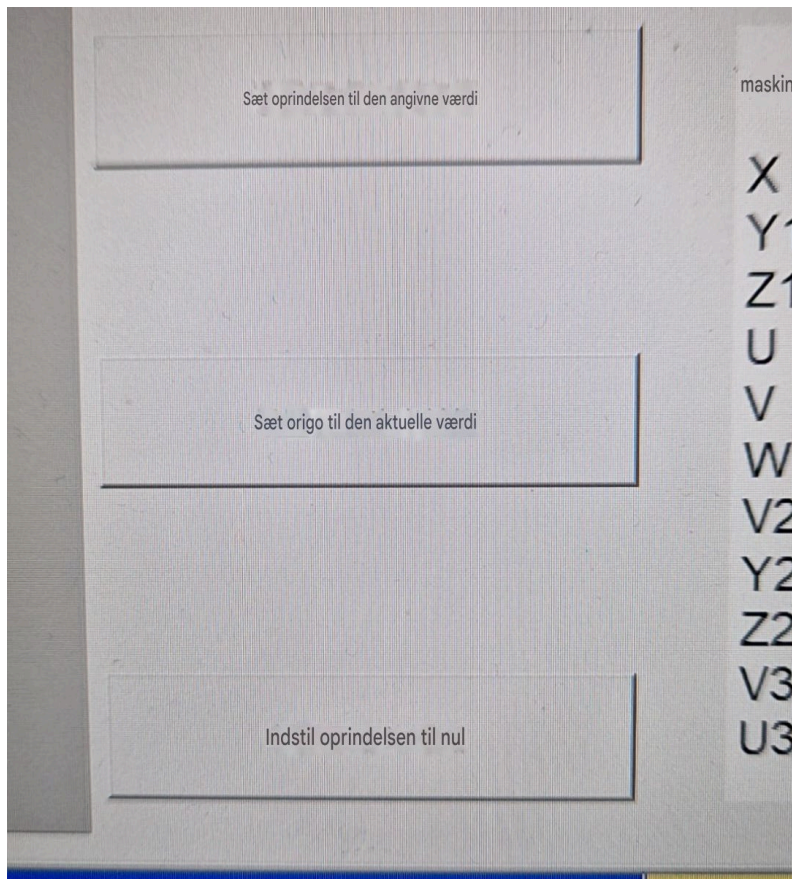


2. Move it according to the picture shown.



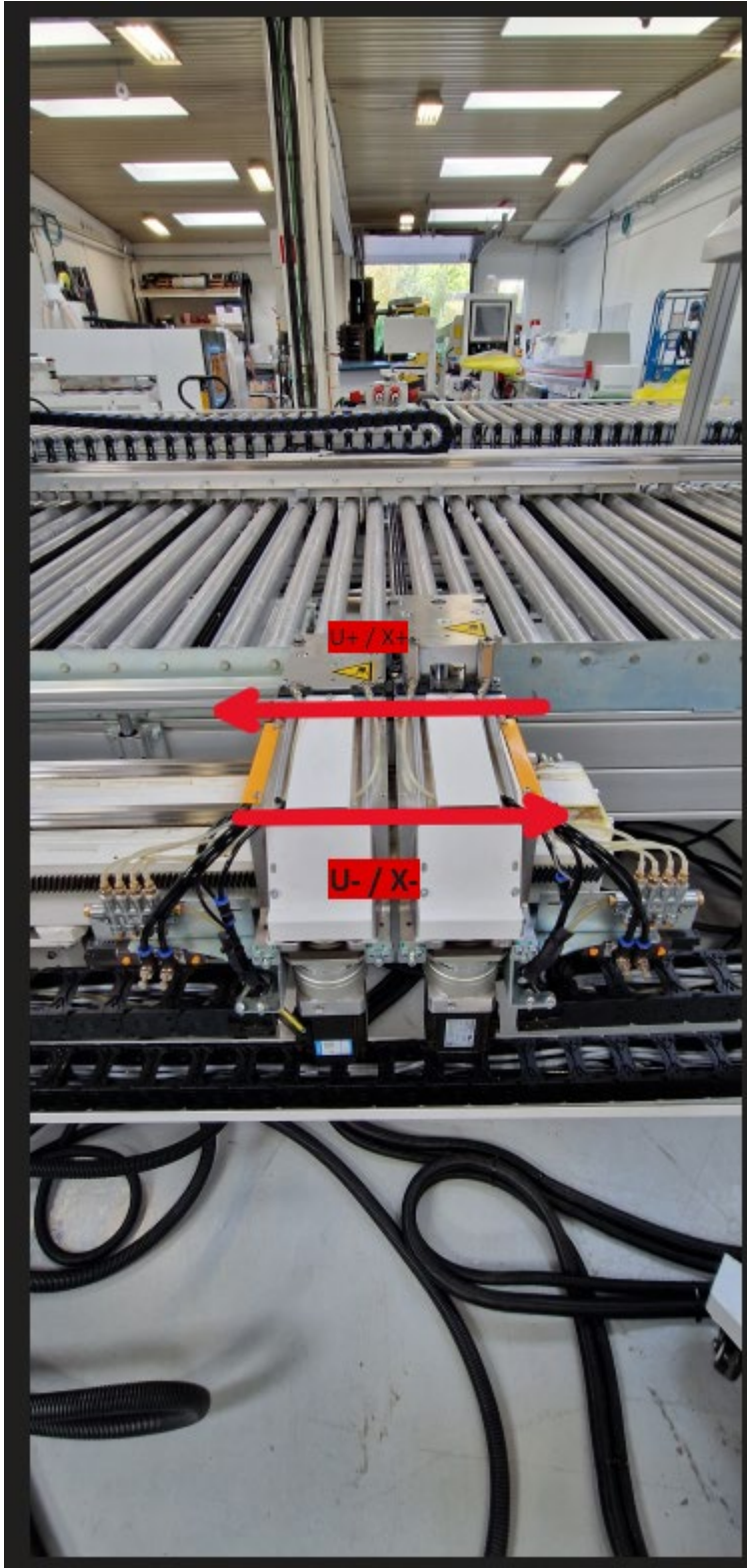
3. When the clamps is positioned, go into the Absolut Origin, set the current mode to HOME, and zero out the X and U axis by clicking the middle one (not shown text) but this is meaning set the zero position to the current value.





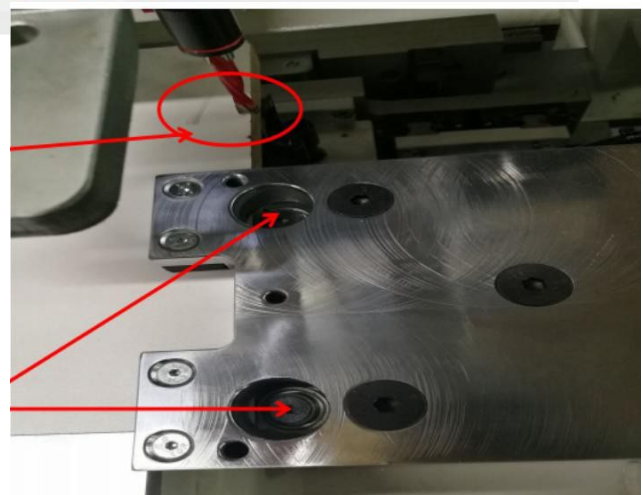
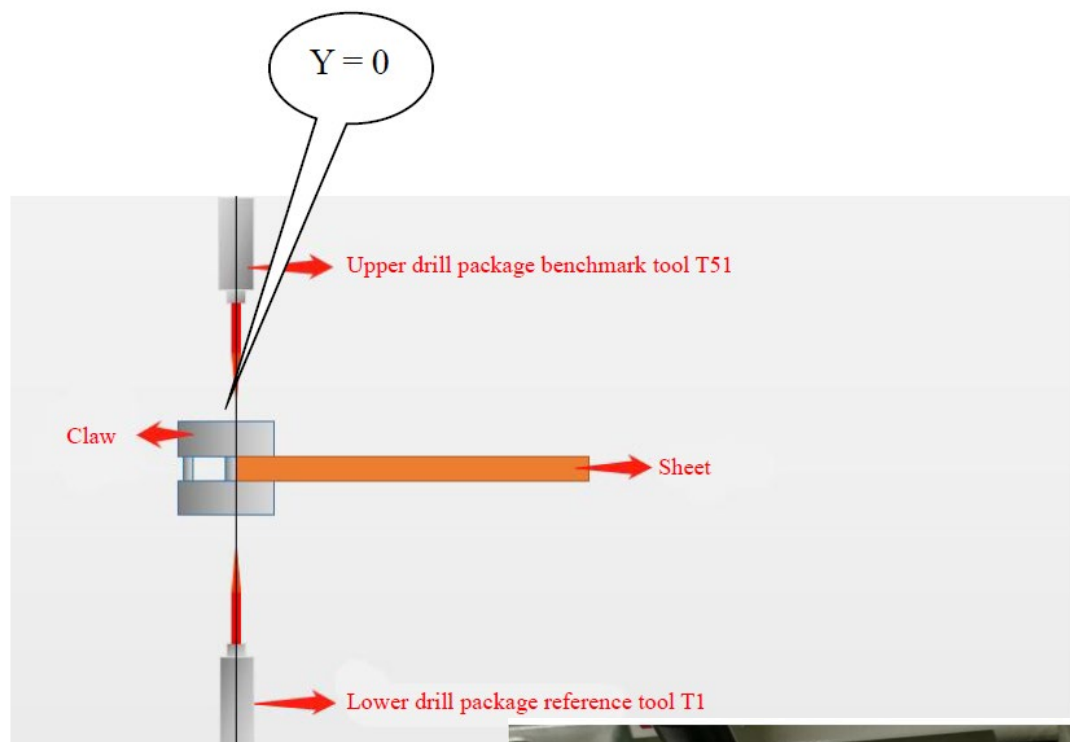


4. Now do the same the U axis, so it goes to the same position.
5. Zero out the U position, using the same method.



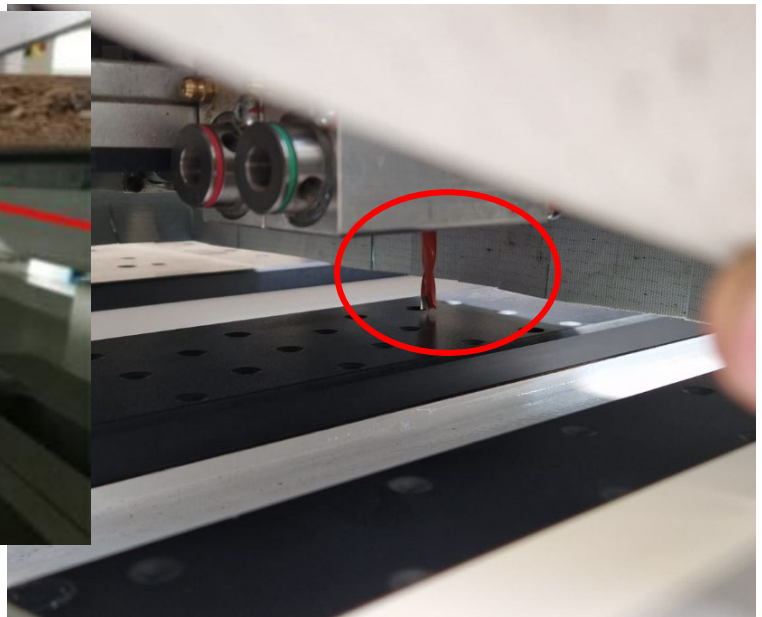
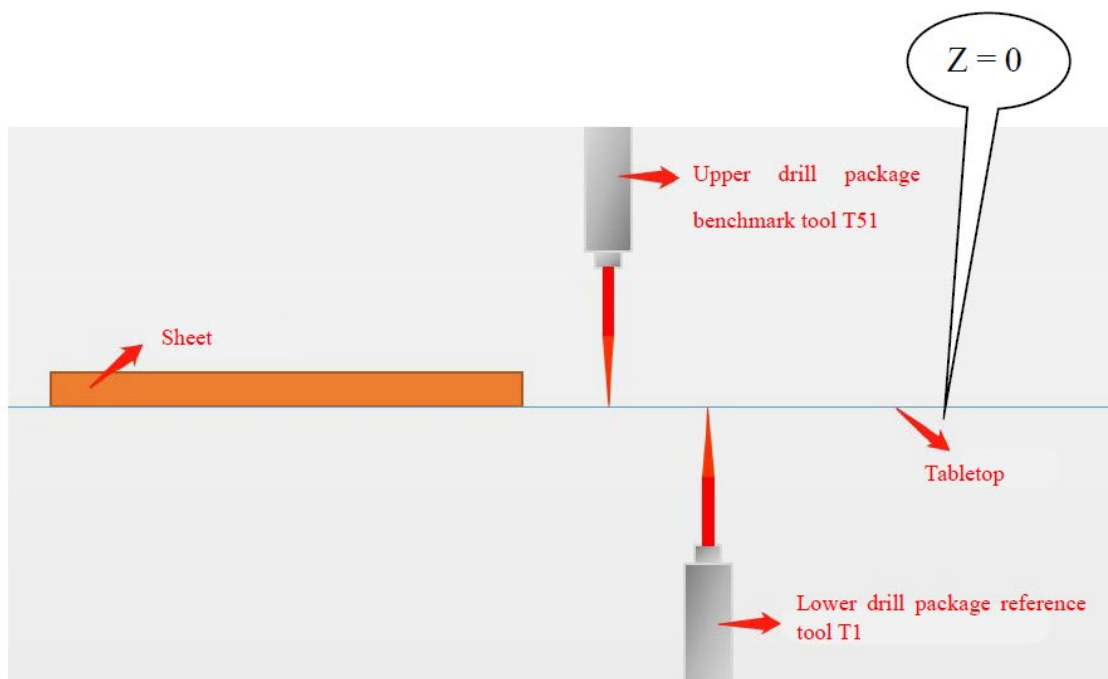
## The origin of the drill box in the Y axis direction

1. Origin coordinate setting of the Y1 axis on the upper drill box: Take out the reference tool T51 of the upper rotary box 1, and move the upper drill box Y1 axis so that the tool nose is just aligned with the contact surface between the side of the claw sheet and plate and the side gear of the claw.
2. Origin coordinate setting of the Y2 axis on the upper drill box: Take out the reference tool T151 of the upper rotary box 2 and move the upper drill box Y2 axis so that the tool nose is just aligned with the contact surface between the side of the claw sheet and plate and the side gear of the claw.
3. Origin coordinate setting of the W axis on the lower drill box: Take in the reference tool T1 of the lower rotary box, move the W axis on the lower drill box so that the tool nose is just aligned with the contact surface between the side of the claw sheet and plate and the side gear of the claw.



## The origin of the drill box in the Z axis direction

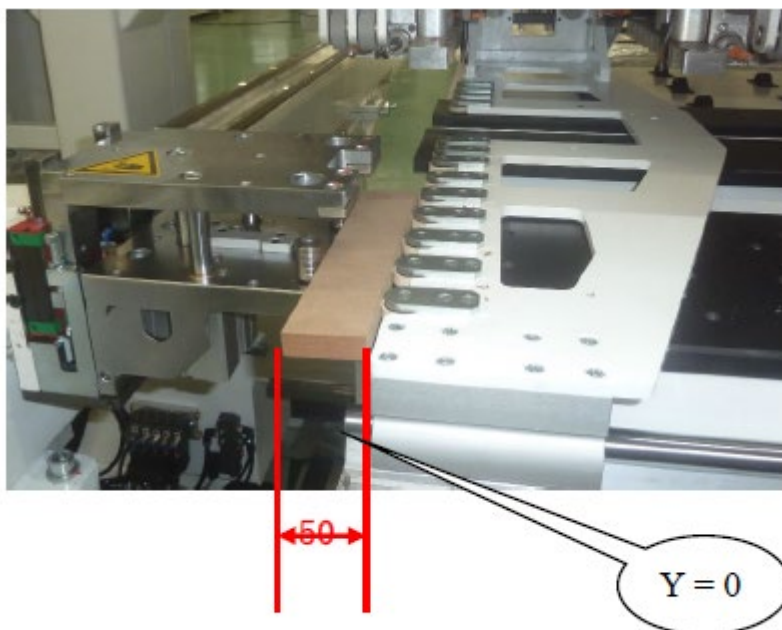
1. Coordinate setting of the Z1 axis on the upper drill box: Take out the reference tool T51, and move the Z1 axis so that the tool nose just touches the tabletop.
2. The coordinate setting of the Z2 axis of the upper drill box: Punch down the reference tool T151 and move the Z2 axis so that the tool nose just touches the tabletop.
3. Coordinate setting of the Z axis of the lower drill box: Take out the reference tool T1, and move the Z axis so that the tool nose just touches the tabletop.



## Origin of side crosspiece V axis

1. Move the side crosspiece V axis laterally to the position close to the sheet;
2. The distance between the positioning post and the V axis side crosspiece, that is, the width of the sheet;
3. If the distance (or width of the board) is 50mm:
4. In the position of V axis G54, fill in -50.

If the side crosspiece is on the other side of the sheet, fill in 50 at the position of V axis G54.



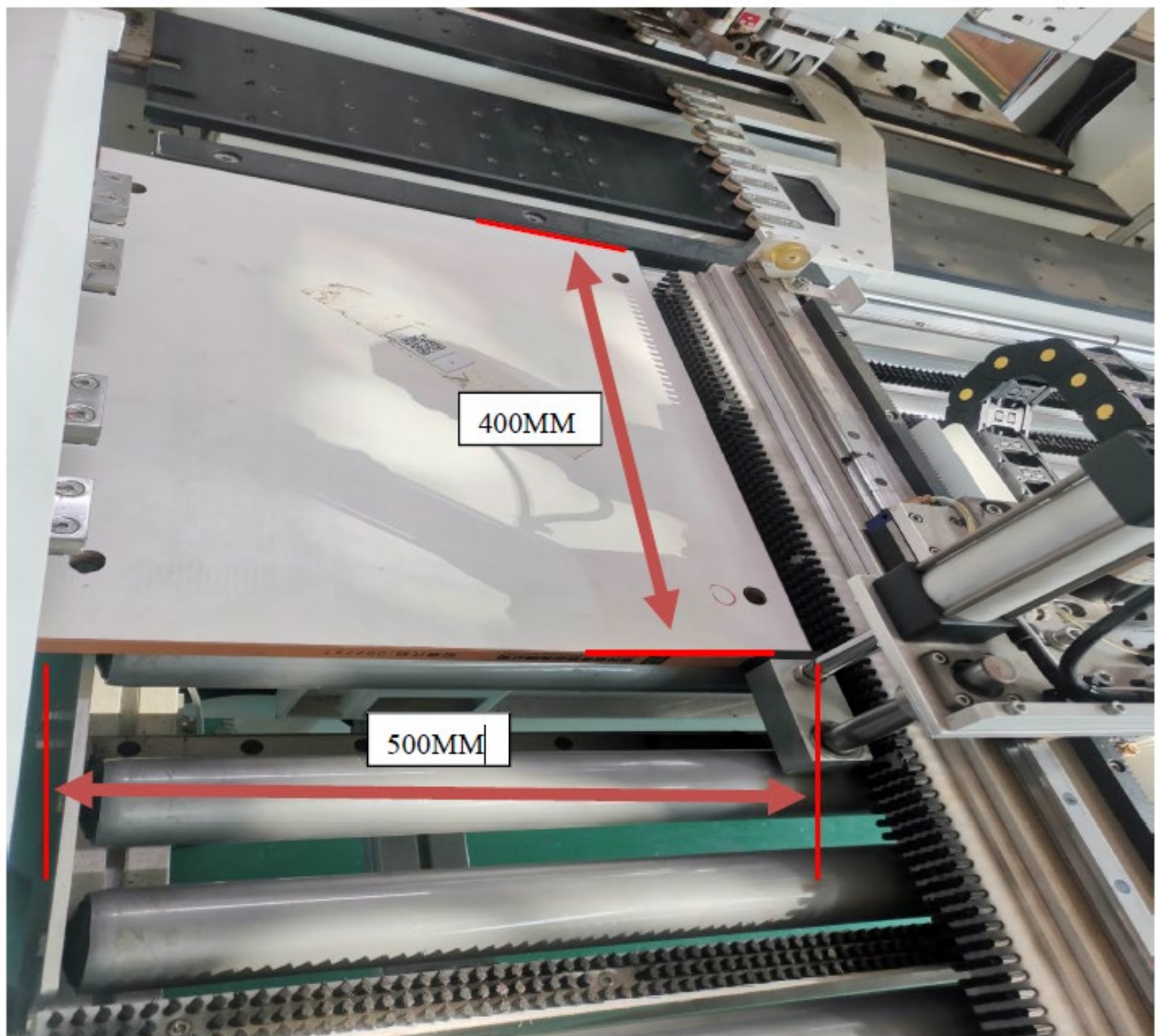


## Original points of side thrust U3-axis and V3-axis

1. Move the side crosspiece V3 axis laterally to the position close to the sheet;
2. The distance between the positioning post and the V3 axis side crosspiece, that is, the width of the sheet;
3. If the distance (or width of the board) is 500mm:
4. In the position of V3 axis G54, fill in -500.

If the side crosspiece is on the other side of the sheet, fill in 500 at the position of V axis G54.

5. Move the side gear U3 axis vertically to a position close to the sheet;
6. The distance between the positioning post and the U3 axis side crosspiece, that is, the width of the sheet;
7. If the distance (or plate length) is 400mm:
8. Fill in - 400 in the G54 position of U3 axis. If the side crosspiece is on the other side of the sheet, fill in 400 at the position of U3 axis G54. 500MM



Go to parameter 3418 (Normal mode and Engineering mode) and change it to 0  
Code again here is 520.



# Test run and adjusting

Perform a test run for the machine, by using the PCDrillCAD program.

## Machining accuracy confirmation

1. To adjust the accuracy, first adjust the accuracy of the reference tool, and then adjust the accuracy of the horizontal tool and the spindle;

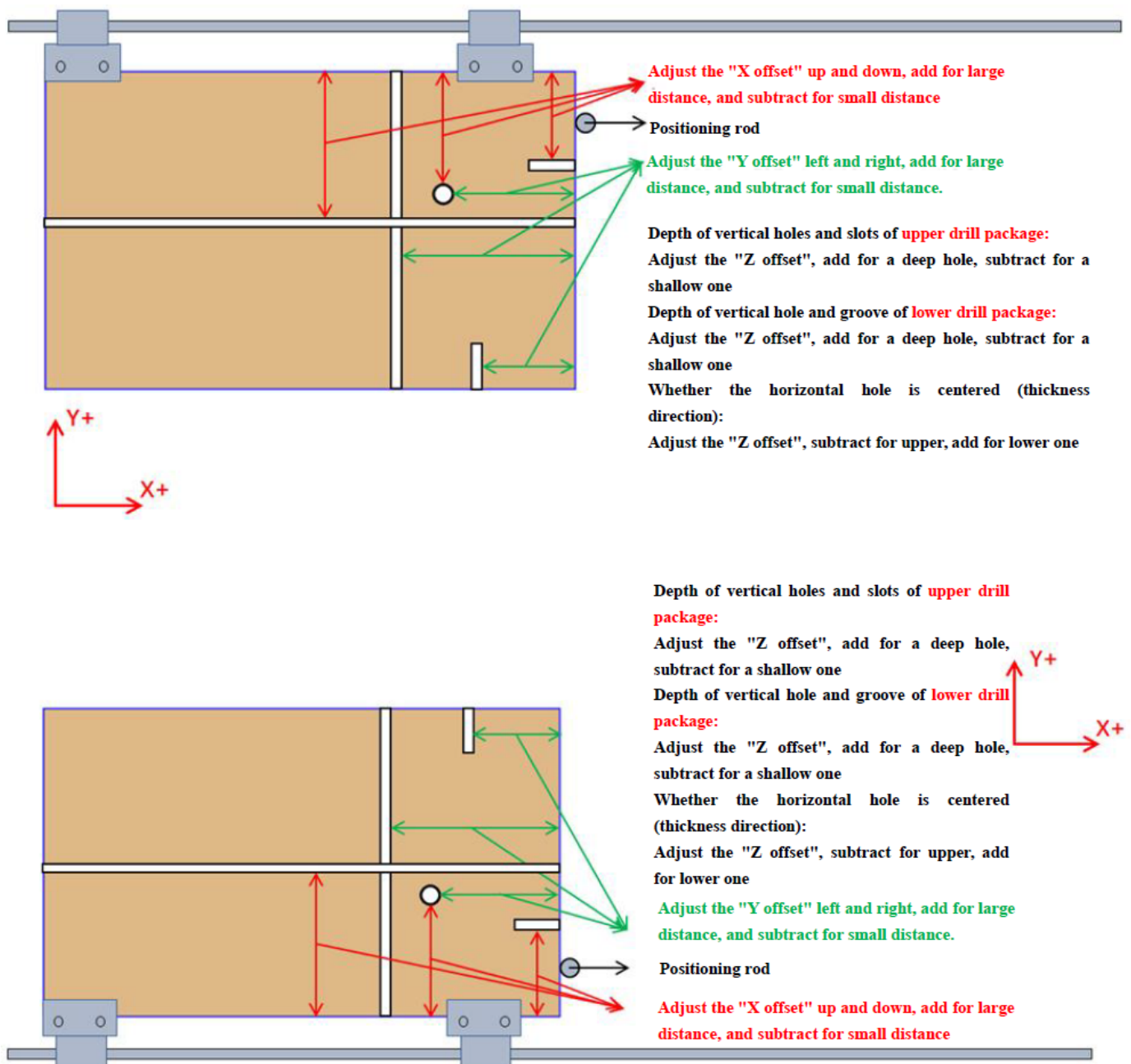
2. Accuracy overall offset, adjust G54 coordinates and parameters 4711/4714/4720;

Parameter 4711: reference point X of lower drill box 1;

Parameter 4714, reference point X of the upper drill box 1.

Parameter 4720, reference point X of the upper drill box 2.

If a single tool is not allowed, adjust the tool data sheet.

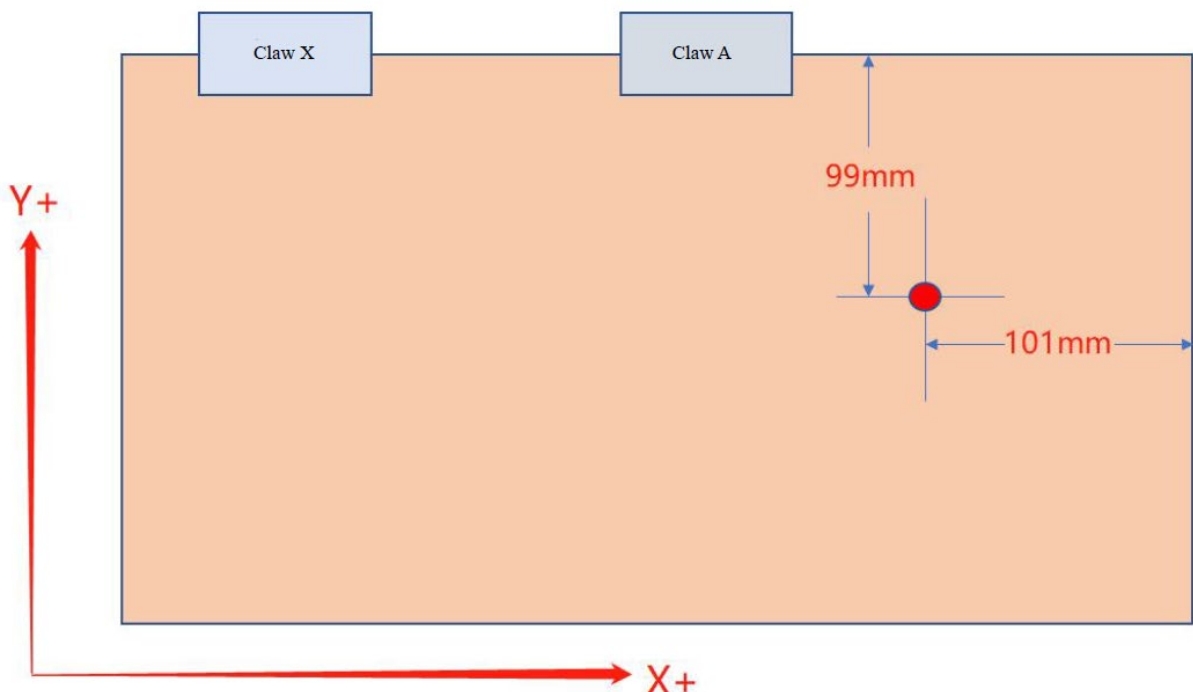


## Vertical hole X axis, Y axis position adjustment

If the positional accuracy of the X axis and Y axis of the vertical hole is not accurate, it is only necessary to adjust the accuracy of the reference tool (T1/T51/T151) of each drill box; other vertical tools do not need to be adjusted. Because the spacing between other vertical tools and the reference tool is an integral multiple of 32 mm, which has been set in the tool list, so the drilling precision of the reference tool is accurate, and the drilling precision of other vertical tools is accurate;

Note: If the precision of the reference tool is accurate, and the precision of other vertical tools is inaccurate at this time, you need to modify the X offset and Y offset of the tool data list according to the actual situation.

1. It is expected to drill a horizontal hole with a depth of 100mm at a distance of 100mm away from the gripper side and 100mm from the sheet head;
2. After the actual processing, it is found that the accuracy is not accurate, the X-direction is 1mm more, and the Y direction is 1mm less, as shown below:



Y direction: Modify the external coordinate offset in the Y direction from 0 to -1, as



shown in the figure below:

| External coordinate offset |       | G54P1(G54)  | G54P2(G55) | Machinery coordinate       |       |
|----------------------------|-------|-------------|------------|----------------------------|-------|
| X                          | 0.000 | X 1684.380  | X 0.000    | X                          | 0.000 |
| Y                          | 0.000 | Y -396.213  | Y 0.000    | Y                          | 0.000 |
| Z                          | 0.000 | Z -87.793   | Z 0.000    | Z                          | 0.000 |
| A                          | 0.000 | A -2526.900 | A 0.000    | A                          | 0.000 |
| U                          | 0.000 | U -1200.000 | U 0.000    | U                          | 0.000 |
| G54P3(G56)                 |       | G54P4(G57)  | G54P5(G58) | Relative coordinate        |       |
| X                          | 0.000 | X 0.000     | X 0.000    | X                          | 0.000 |
| Y                          | 0.000 | Y 0.000     | Y 0.000    | Y                          | 0.000 |
| Z                          | 0.000 | Z 0.000     | Z 0.000    | Z                          | 0.000 |
| A                          | 0.000 | A 0.000     | A 0.000    | A                          | 0.000 |
| U                          | 0.000 | U 0.000     | U 0.000    | U                          | 0.000 |
|                            |       |             |            | Auxiliary point coordinate |       |
|                            |       |             |            | X                          | 0.000 |
|                            |       |             |            | Y                          | 0.000 |
|                            |       |             |            | Z                          | 0.000 |

| External coordinate offset |        | G54P1(G54)  | G54P2(G55) | Machinery coordinate       |       |
|----------------------------|--------|-------------|------------|----------------------------|-------|
| X                          | 0.000  | X 1684.380  | X 0.000    | X                          | 0.000 |
| Y                          | -1.000 | Y -396.213  | Y 0.000    | Y                          | 0.000 |
| Z                          | 0.000  | Z -87.793   | Z 0.000    | Z                          | 0.000 |
| A                          | 0.000  | A -2526.900 | A 0.000    | A                          | 0.000 |
| U                          | 0.000  | U -1200.000 | U 0.000    | U                          | 0.000 |
| G54P3(G56)                 |        | G54P4(G57)  | G54P5(G58) | Relative coordinate        |       |
| X                          | 0.000  | X 0.000     | X 0.000    | X                          | 0.000 |
| Y                          | 0.000  | Y 0.000     | Y 0.000    | Y                          | 0.000 |
| Z                          | 0.000  | Z 0.000     | Z 0.000    | Z                          | 0.000 |
| A                          | 0.000  | A 0.000     | A 0.000    | A                          | 0.000 |
| U                          | 0.000  | U 0.000     | U 0.000    | U                          | 0.000 |
|                            |        |             |            | Auxiliary point coordinate |       |
|                            |        |             |            | X                          | 0.000 |
|                            |        |             |            | Y                          | 0.000 |
|                            |        |             |            | Z                          | 0.000 |

X direction: Modify parameter 4714, change 166900 (representing 166.9 mm) to 165900 (representing 165.9 mm).

| No.    | Parameter number | Parameter description                                                                                                                 | Value    |
|--------|------------------|---------------------------------------------------------------------------------------------------------------------------------------|----------|
| Cutter |                  |                                                                                                                                       |          |
| 1      | 4526             | Distance from reference tool to front edge of front platen                                                                            | 0        |
| 2      | 4711             | Lower drill package 1 reference point X                                                                                               | 294900   |
| 3      | 4714             | Upper drill package 1 reference point X                                                                                               | 166900   |
| 4      | 4720             | Upper drill package 2 reference point X                                                                                               | 166900   |
| 5      | 4900             | Feeding speed                                                                                                                         | 0        |
| 6      | 4906             | Saw blade cutting speed (mm/min)/m                                                                                                    | 0        |
| 7      | 4907             | Spindle cutting rate (mm/min)                                                                                                         | 0        |
| 8      | 4958             | The "through" technique is changed to up/down parts (bit control hole +1 slot +2 milling type +4).                                    | 0        |
| 9      | 4959             | Overlapping amount (length: positive, percentage: negative) when "through technique" is changed to "up/down technique"                | 0        |
| 10     | 4963             | Combine "through" and "vertical" techniques (No =0, drilling +1, slotting +2, milling type +4, turning front +128, turning back +256) | 0        |
| 11     | 4964             | Merging through the vertical process, sheet and plate will not be merged if the thickness reaches this upper limit                    | 0        |
| 12     | 4965             | Merging through the vertical process, add additional depth                                                                            | 0        |
| 13     | 4966             | "Through vertical" technique face changing (no =0, drilling +1, slotting +2, milling type +4, turning front +128, turning back +256)  | 0        |
| Claw   |                  |                                                                                                                                       |          |
| 1      | 4520             | Claw X interference distance                                                                                                          | 10000    |
| 2      | 4560             | Claw Y interference distance                                                                                                          | 15000    |
| 3      | 4521             | Claw X length                                                                                                                         | 145000   |
| 4      | 4522             | Claw Y length                                                                                                                         | 33000    |
| 5      | 4523             | Claw Z thickness                                                                                                                      | 30000    |
| 6      | 4524             | Minimum clamping distance in the X-direction of the claw                                                                              | 30000    |
| 7      | 4561             | Minimum clamping distance in the Y-direction of the claw                                                                              | 0        |
| 8      | 4529             | Claw is at the G54 positive limit position                                                                                            | 2330000  |
| 9      | 4530             | Claw is in the negative limit position of G54                                                                                         | -1800000 |
| 10     | 4556             | Allow a single claw to feed plate (No=0, Yes=1)                                                                                       | 0        |

## Vertical hole Z axis depth adjustment

Take the upper drill box 1 as an example, and its reference tool is T51 with a diameter of 8 mm.

Use the reference tool T51 to drill a 10 mm deep hole, the actual measurement is only 9 mm, which is 1 mm shallower; then modify the external coordinate offset, modify the external coordinate offset of Z from 0 to -1, as shown in the following figure:

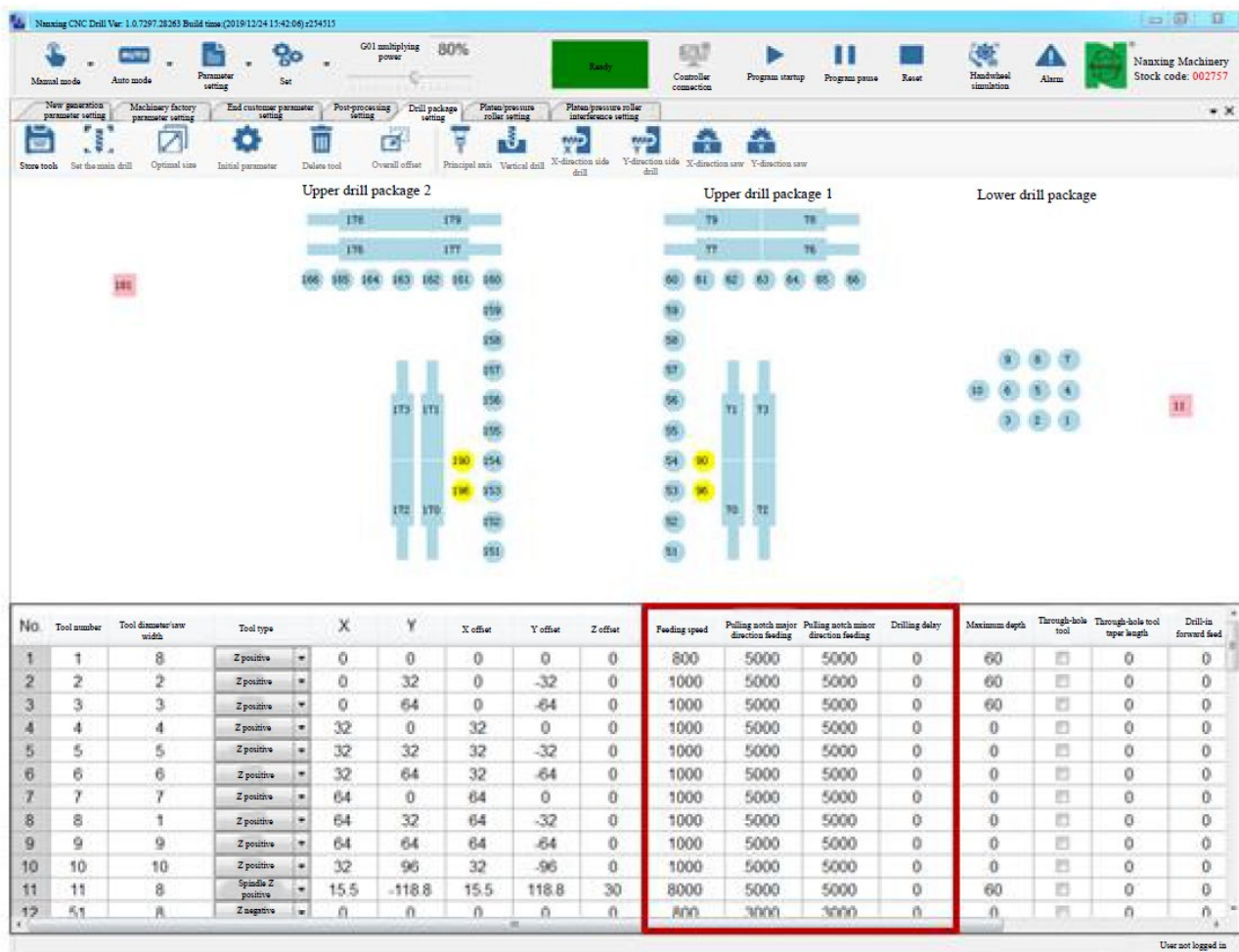
Operation path: Machine setting - workpiece coordinate system

| External coordinate offset |       | G54P1(G54)  | G54P2(G55) | Machinery coordinate |                            |
|----------------------------|-------|-------------|------------|----------------------|----------------------------|
| X                          | 0.000 | X 1684.380  | X 0.000    | X                    | 0.000                      |
| Y                          | 0.000 | Y -396.213  | Y 0.000    | Y                    | 0.000                      |
| Z                          | 0.000 | Z -87.793   | Z 0.000    | Z                    | 0.000                      |
| A                          | 0.000 | A -2526.900 | A 0.000    | A                    | 0.000                      |
| U                          | 0.000 | U -1200.000 | U 0.000    | U                    | 0.000                      |
| G54P3(G56)                 |       | G54P4(G57)  | G54P5(G58) | Relative coordinate  |                            |
| X                          | 0.000 | X 0.000     | X 0.000    | X                    | 0.000                      |
| Y                          | 0.000 | Y 0.000     | Y 0.000    | Y                    | 0.000                      |
| Z                          | 0.000 | Z 0.000     | Z 0.000    | Z                    | 0.000                      |
| A                          | 0.000 | A 0.000     | A 0.000    | A                    | 0.000                      |
| U                          | 0.000 | U 0.000     | U 0.000    | U                    | 0.000                      |
|                            |       |             |            |                      | Auxiliary point coordinate |
|                            |       |             |            |                      | X 0.000                    |
|                            |       |             |            |                      | Y 0.000                    |
|                            |       |             |            |                      | Z 0.000                    |

| External coordinate offset |        | G54P1(G54)  | G54P2(G55) | Machinery coordinate |                            |
|----------------------------|--------|-------------|------------|----------------------|----------------------------|
| X                          | 0.000  | X 1684.380  | X 0.000    | X                    | 0.000                      |
| Y                          | 0.000  | Y -396.213  | Y 0.000    | Y                    | 0.000                      |
| Z                          | -1.000 | Z -87.793   | Z 0.000    | Z                    | 0.000                      |
| A                          | 0.000  | A -2526.900 | A 0.000    | A                    | 0.000                      |
| U                          | 0.000  | U -1200.000 | U 0.000    | U                    | 0.000                      |
| G54P3(G56)                 |        | G54P4(G57)  | G54P5(G58) | Relative coordinate  |                            |
| X                          | 0.000  | X 0.000     | X 0.000    | X                    | 0.000                      |
| Y                          | 0.000  | Y 0.000     | Y 0.000    | Y                    | 0.000                      |
| Z                          | 0.000  | Z 0.000     | Z 0.000    | Z                    | 0.000                      |
| A                          | 0.000  | A 0.000     | A 0.000    | A                    | 0.000                      |
| U                          | 0.000  | U 0.000     | U 0.000    | U                    | 0.000                      |
|                            |        |             |            |                      | Auxiliary point coordinate |
|                            |        |             |            |                      | X 0.000                    |
|                            |        |             |            |                      | Y 0.000                    |
|                            |        |             |            |                      | Z 0.000                    |

Although the tool noses of all vertical tools (including T51) are on the same horizontal plane, the tool diameters are different. Drill at the same speed. If the air pressure is not particularly sufficient, a tool with a larger tool diameter (such as a 20 mm knife) will drill shallower than a tool with a smaller tool diameter (such as a 5 mm knife).

Therefore, the feeding speed shall be modified according to the actual situation, or the hole bottom delay shall be set.



Notice: It is also possible that the tool noses of other tools and the reference tool are not on the same level. In this case, the Z offset of the tool data sheet needs to be modified according to the actual situation;



## Side hole accuracy adjustment

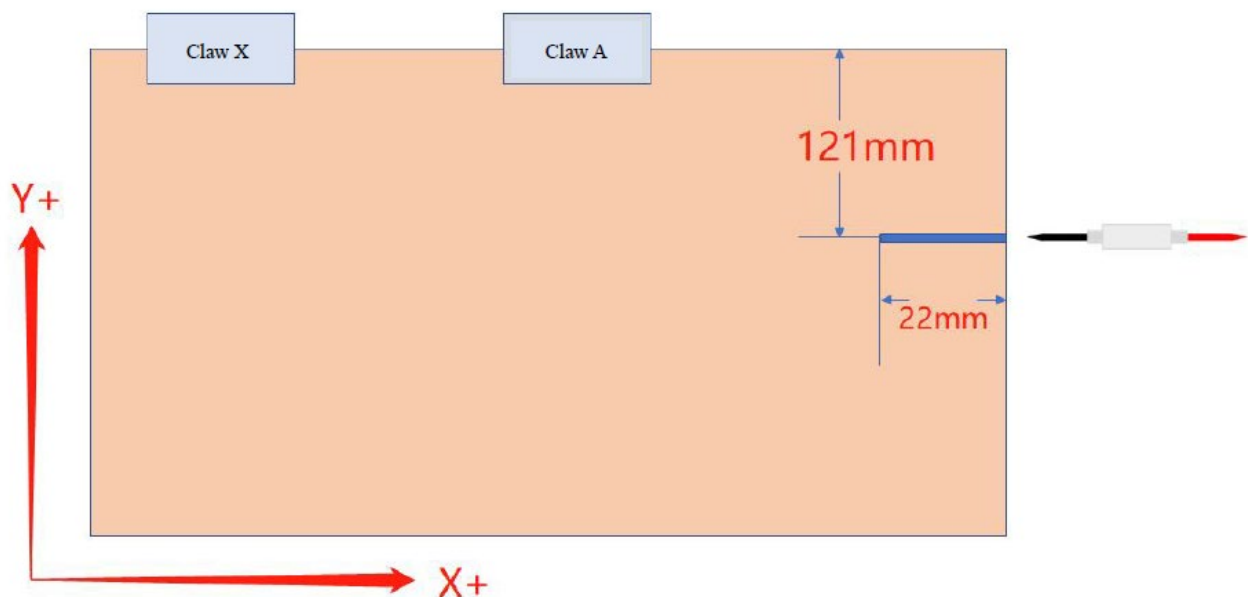
The horizontal tool diameter is calculated in 8mm here;

Accuracy adjustment of X-direction side hole

1. It is expected to drill a horizontal hole with a depth of 20mm at a distance of 120mm from the gripper side;

2. After the actual processing, it is found that the accuracy is not accurate, the

X-direction is 2mm deep, and the Y-direction is 1mm more, as shown below:



3. Enter the tool data list and find the corresponding tool (this example shows the horizontal hole in the positive X direction, as shown below):

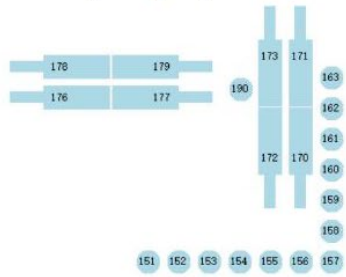
4. At this time, we hope that the corresponding tool moves 2mm in the X+ direction and 1mm in the Y+ direction, so that the hole position is correct;

5. Modify the X offset and Y offset of the tool list, subtract 2 from the X offset, and add 1 to the Y offset;

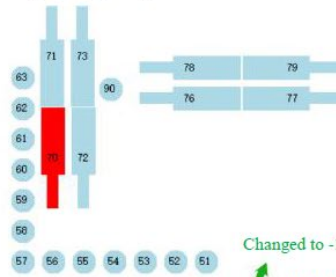
6. Both T70 and T72 are horizontal tools in the positive X-direction, but the difference in the Y-direction is 32mm, which can be modified together.



Upper drill package 2



Upper drill package 1



Lower drill package



Changed to 52.6

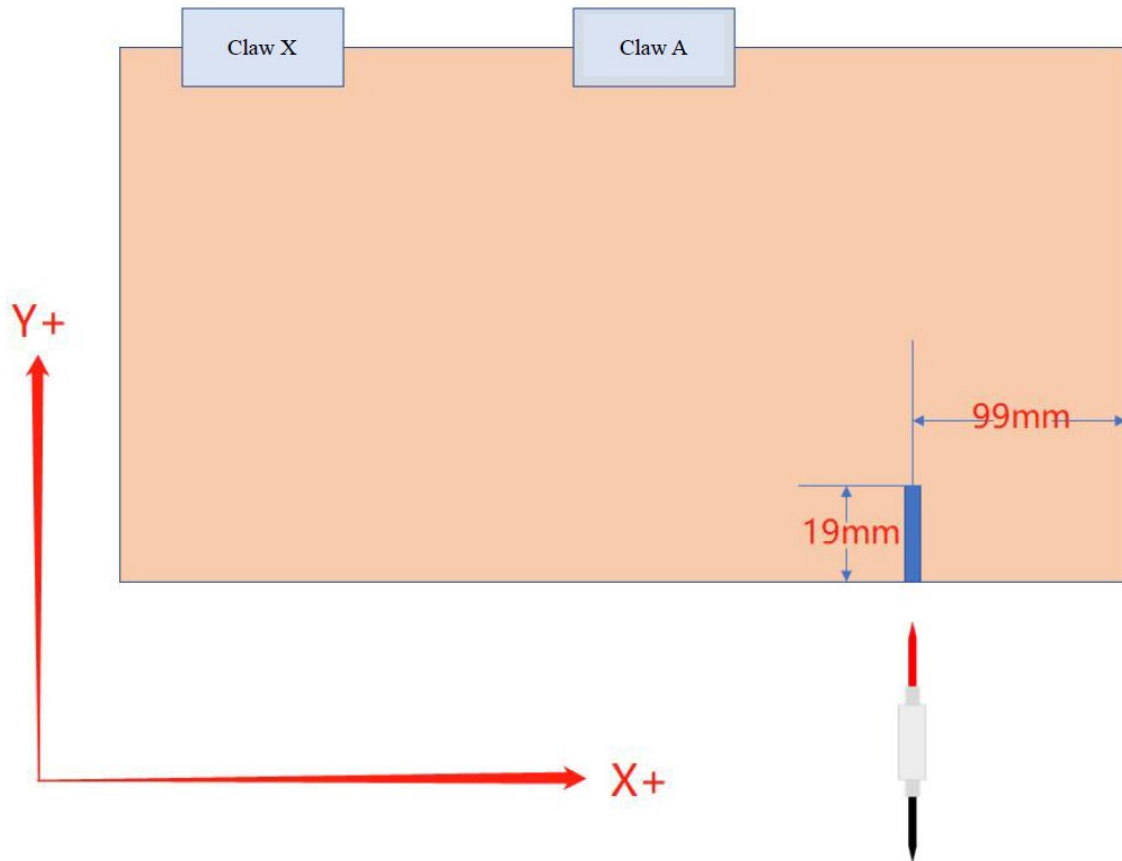
Changed to -159.2

Changed to -127.2

| No. | Tool number | Tool diameter/saw width | Tool type  | X    | Y      | X offset | Y offset | Z offset | Feeding speed | Drilling delay | Maximum depth |
|-----|-------------|-------------------------|------------|------|--------|----------|----------|----------|---------------|----------------|---------------|
| 25  | 70          | 8                       | X positive | 54.6 | 160.2  | 54.6     | -160.2   | -11.5    | 5000          | 0.1            | 60            |
| 26  | 71          | 8                       | X negative | 266  | 160.6  | 266      | -160.6   | -11.6    | 5000          | 0.1            | 60            |
| 27  | 72          | 8                       | X positive | 54.6 | 128.2  | 54.6     | -128.2   | -11.5    | 5000          | 0.1            | 60            |
| 28  | 73          | 8                       | X negative | 266  | 128.6  | 266      | -128.6   | -11.6    | 5000          | 0.1            | 60            |
| 29  | 76          | 8                       | Y positive | 170  | 68.8   | 170      | -68.8    | -11.5    | 5000          | 0.2            | 60            |
| 30  | 77          | 8                       | Y negative | 170  | -144.9 | 170      | 144.9    | -11.5    | 1500          | 0.2            | 60            |
| 31  | 78          | 8                       | Y positive | 202  | 68.8   | 202      | -68.8    | -11.5    | 1200          | 0.2            | 60            |
| 32  | 79          | 8                       | Y negative | 202  | -144.9 | 202      | 144.9    | -11.5    | 1500          | 0.2            | 60            |
| 33  | 90          | 8                       | Z negative | 180  | 100    | 180      | -100     | -10      | 5000          | 0              | 30            |
| 34  | 151         | 8                       | Z negative | 0    | 0      | 0        | 0        | 0        | 4000          | 0.1            | 60            |

## Accuracy adjustment of Y-direction side hole

1. It is expected to drill a horizontal hole with a depth of 20mm at a distance of 100mm from the sheet head;
2. After the actual processing, it is found that the accuracy is not accurate, and the X and Y directions are both less than 1mm, as shown in the following figure:

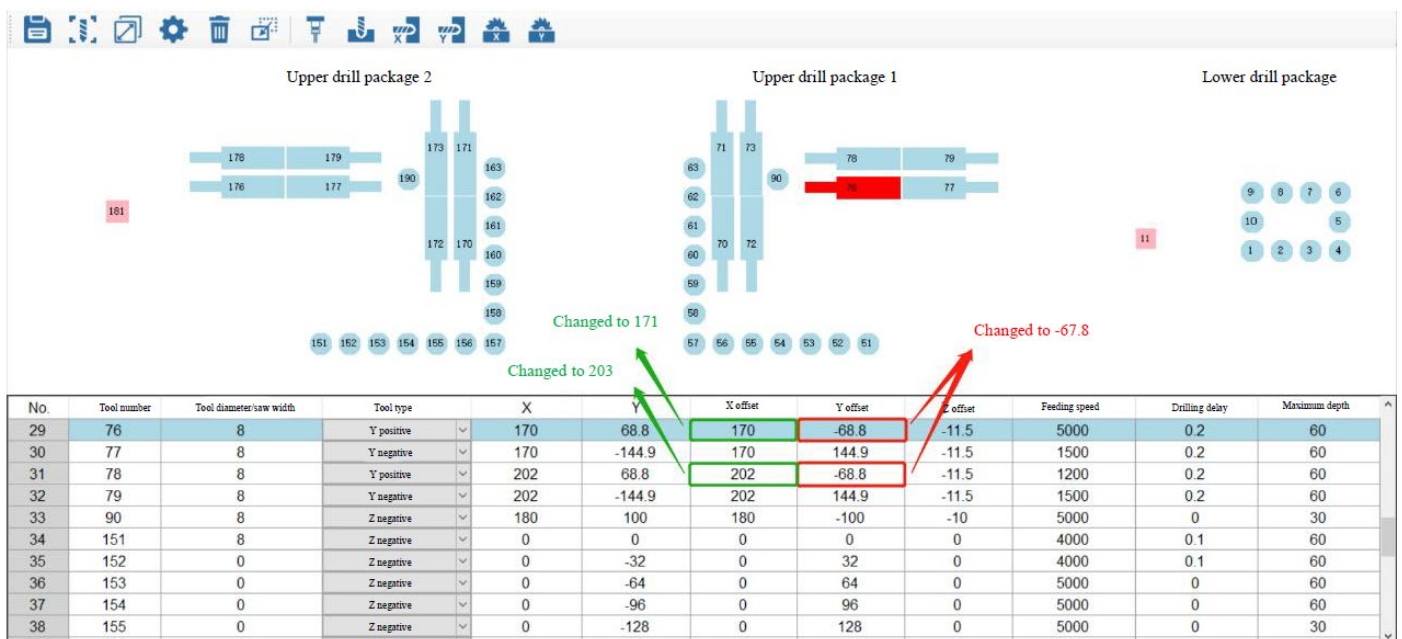


3. Enter the tool data list and find the corresponding tool (this example shows the horizontal hole in the positive Y direction, as shown below):

4. At this time, we hope that the tool moves 1mm in the X- direction and 1mm in the Y+ direction, so that the hole position is correct;

5. Modify the X offset and Y offset of the tool list, subtract 2 from the X offset, and add 1 to the Y offset;

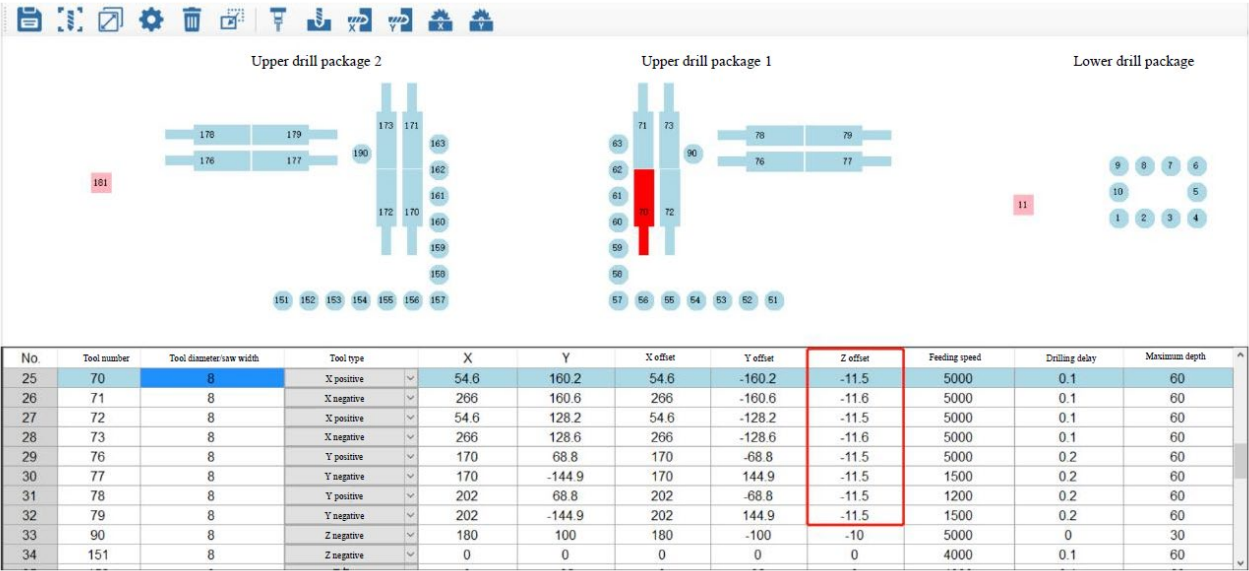
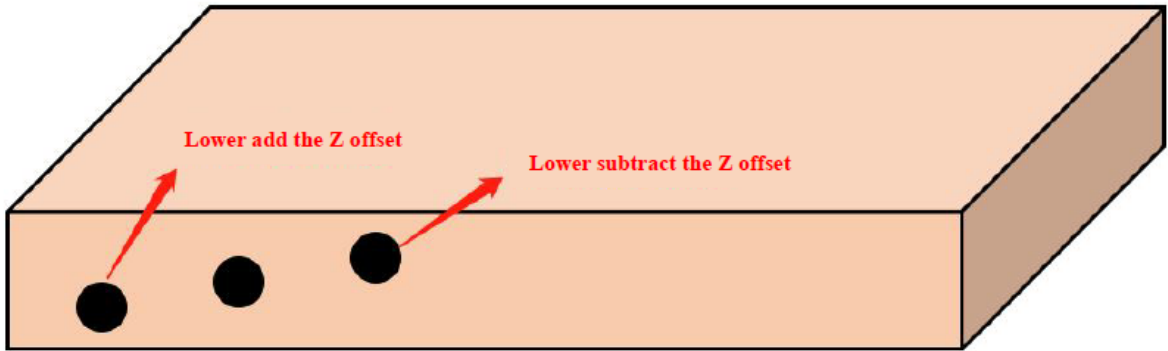
Both T76 and T78 are horizontal tools in the positive Y-direction, but the difference in the X-direction is 32mm, which can be modified together;





# Accuracy adjustment of Z-direction side hole

If the side hole is not in the middle of the sheet, adjust the Z offset of the horizontal tool { as shown in the figure below};

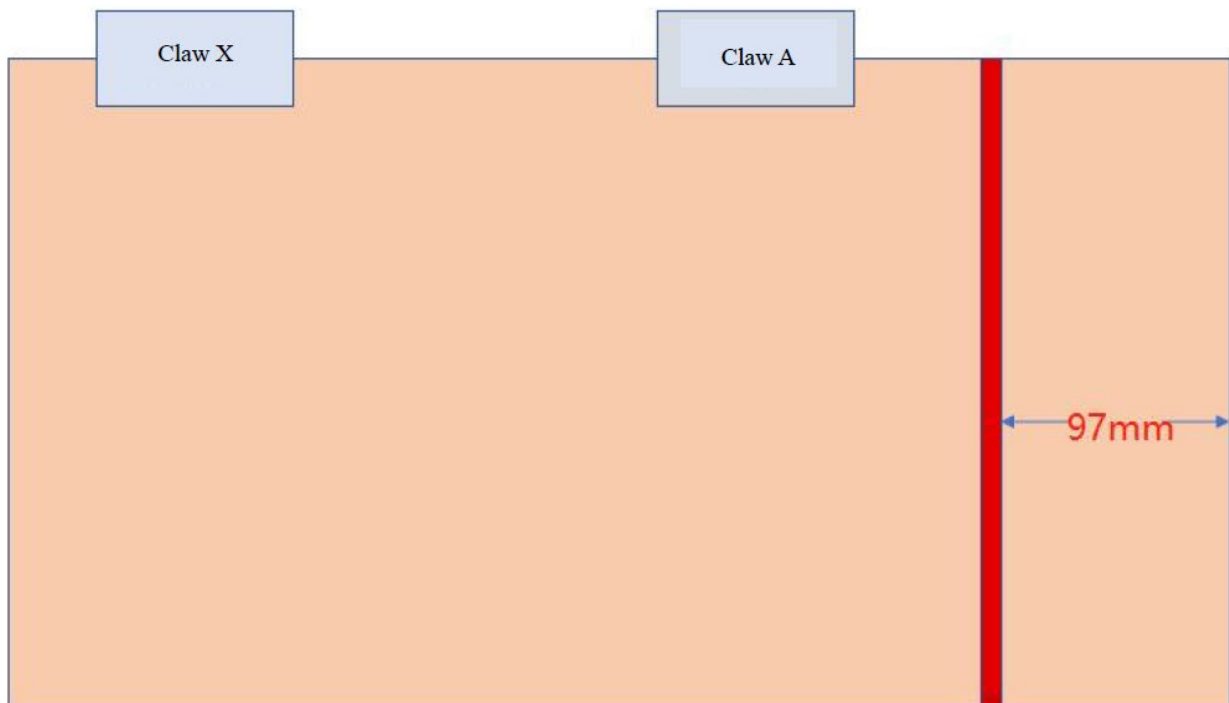


## Spindle accuracy adjustment

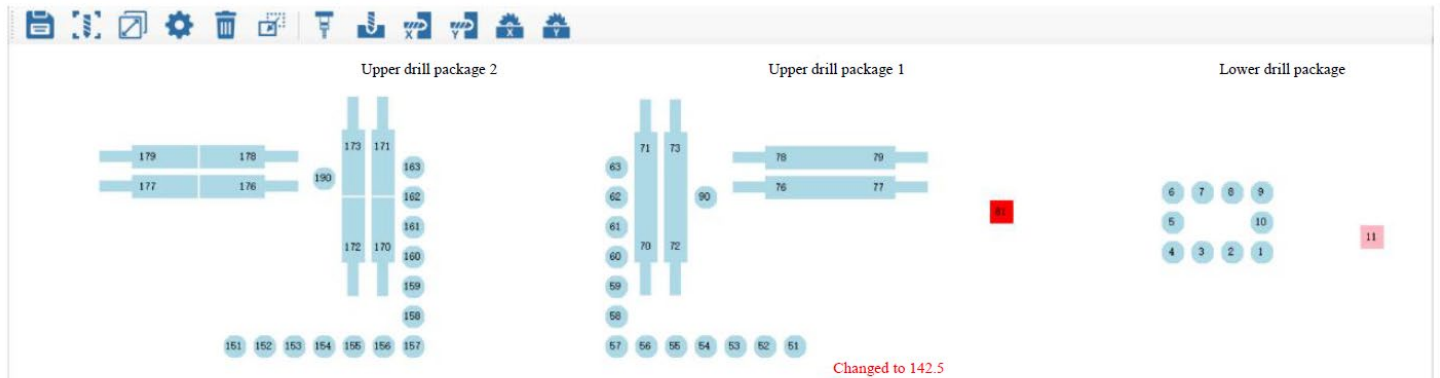
Take the spindle tool diameter of 8mm as an example here;

Accuracy adjustment of X-direction

1. As shown in the figure, a slot is expected to be drawn at a distance of 100mm from the sheet head; in the case of accurate accuracy, the X-direction measured by the caliper should be 96mm (100mm-tool radius 4mm);
2. After the actual processing, it is found that the accuracy is not accurate, and the X-direction is more than 1mm;



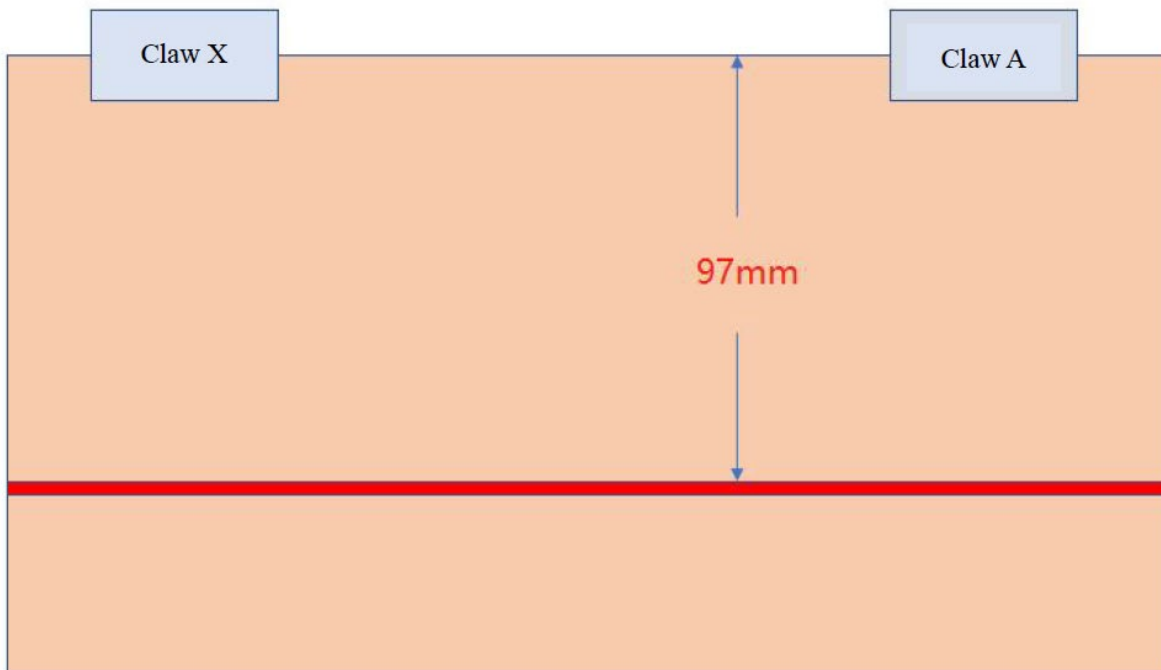
3. Enter the tool data list to find the corresponding spindle (this example shows the spindle Z- as below):
4. At this time, we hope that the corresponding spindle moves 1mm in the X+ direction, so that the slot position is correct;
5. Modify the X offset of the tool sheet and subtract the X offset by 1.



| No. | Tool number | Tool diameter/saw width | Tool type          | X     | Y      | X offset | Y offset | Z offset | Feeding speed | Drilling delay | Maximum depth |
|-----|-------------|-------------------------|--------------------|-------|--------|----------|----------|----------|---------------|----------------|---------------|
| 33  | 81          | 8                       | Spindle Z negative | 143.5 | -222.4 | 143.5    | 222.4    | 40.5     | 5000          | 0              | 0             |
| 34  | 90          | 0                       | Z negative         | 160   | 96.2   | 160      | -96.2    | -10      | 5000          | 0              | 30            |
| 35  | 96          | 0                       | Z negative         | 143.5 | -222.4 | 143.5    | 222.4    | 35.167   | 5000          | 0              | 30            |
| 36  | 151         | 8                       | Z negative         | 0     | 0      | 0        | 0        | 0        | 5000          | 0              | 60            |
| 37  | 152         | 0                       | Z negative         | 0     | -32    | 0        | 32       | 0        | 5000          | 0              | 60            |
| 38  | 153         | 0                       | Z negative         | 0     | -64    | 0        | 64       | 0        | 5000          | 0              | 60            |
| 39  | 154         | 0                       | Z negative         | 0     | -96    | 0        | 96       | 0        | 5000          | 0              | 60            |
| 40  | 155         | 0                       | Z negative         | 0     | -128   | 0        | 128      | 0        | 5000          | 0              | 30            |
| 41  | 156         | 0                       | Z negative         | 0     | -160   | 0        | 160      | 0        | 5000          | 0              | 30            |
| 42  | 157         | 0                       | Z negative         | 0     | -192   | 0        | 192      | 0        | 5000          | 0.2            | 30            |

## Accuracy adjustment of Y-direction

1. As shown in the figure, it is expected to draw a groove at a position 100mm away from the jaw side; in the case of accurate accuracy, the caliper should measure 96mm in the Y-direction (100mm-tool radius 4mm);
2. After the actual processing, it is found that the accuracy is not accurate, and the Y-direction is more than 1mm;

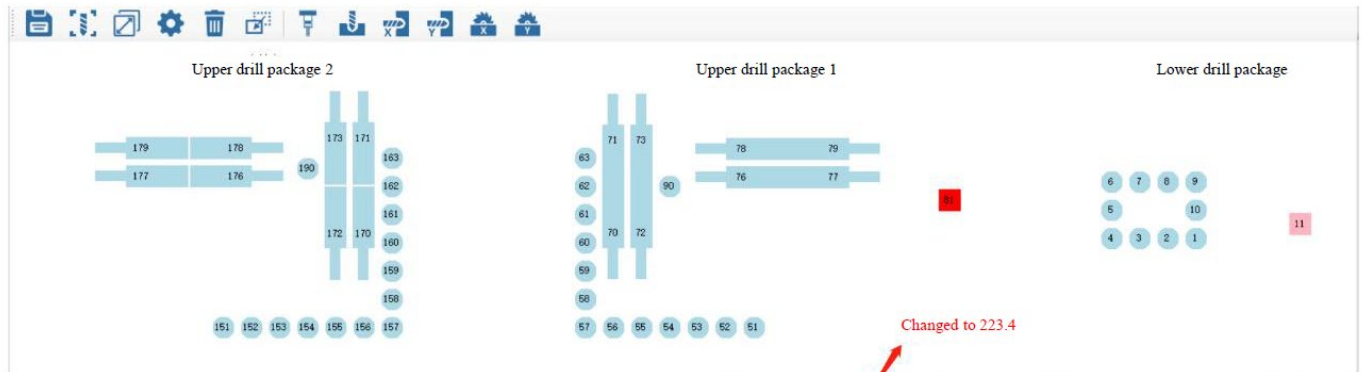




3. Enter the tool data list to find the corresponding spindle (this example shows the spindle Z- as below):

4. At this time, we hope that the corresponding spindle moves 1mm in the Y+ direction, so that the slot position is correct;

Modify the Y offset of the tool sheet and subtract the Y offset by 1.



Upper drill package 2

Upper drill package 1

Lower drill package

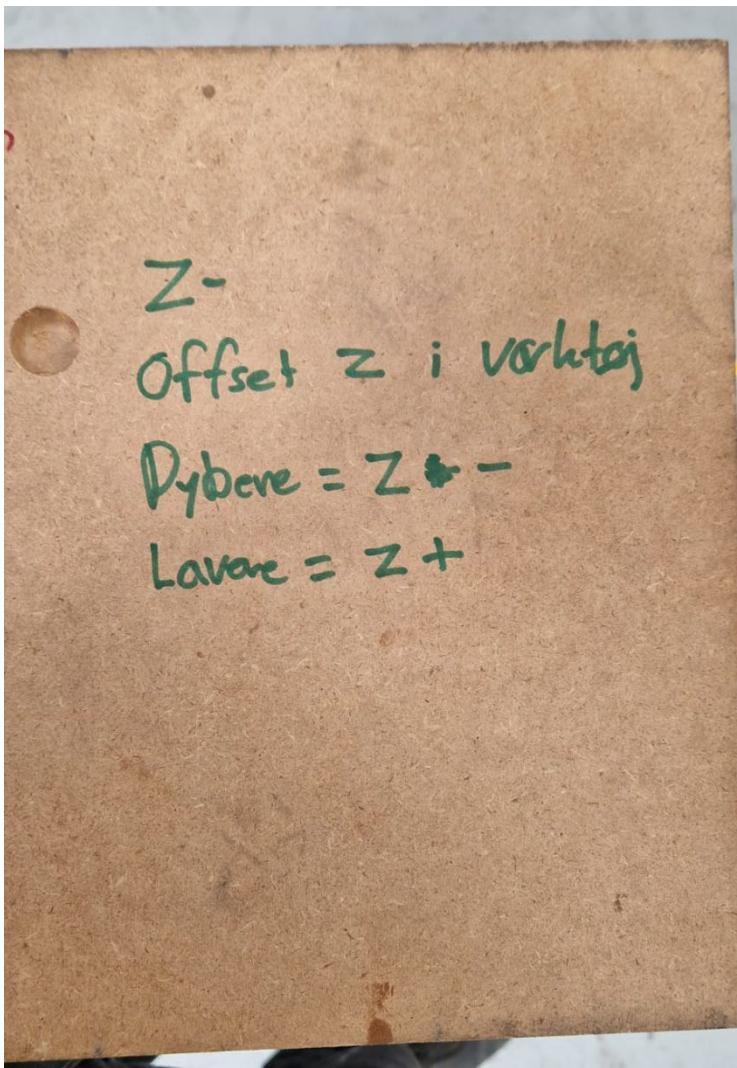
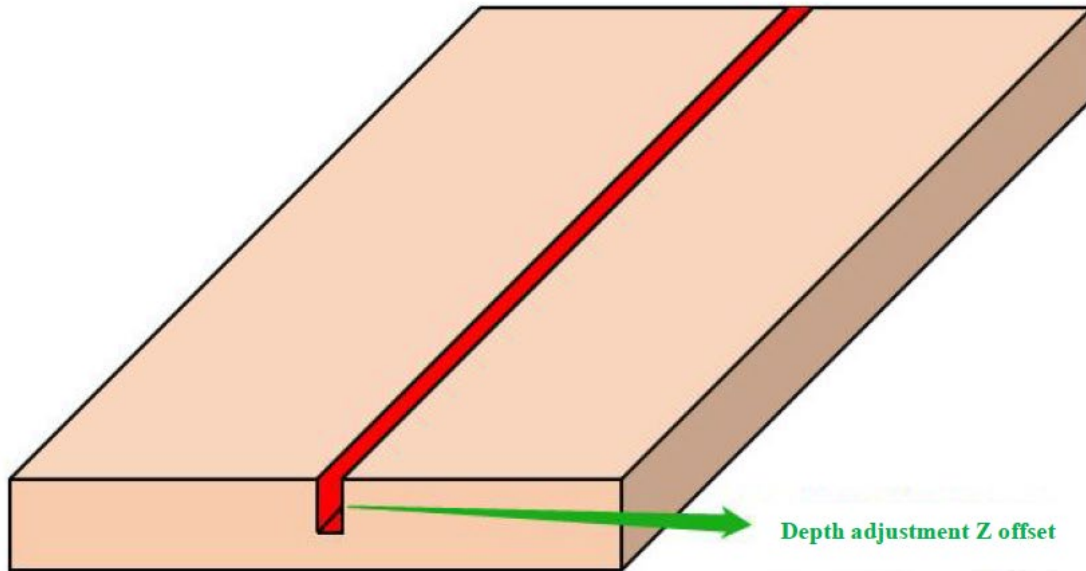
Changed to 223.4

| No. | Tool number | Tool diameter/saw width | Tool type          | X     | Y      | X offset | Y offset | Z offset | Feeding speed | Drilling delay | Maximum depth |
|-----|-------------|-------------------------|--------------------|-------|--------|----------|----------|----------|---------------|----------------|---------------|
| 33  | 81          | 8                       | Spindle Z negative | 143.5 | -222.4 | 143.5    | 222.4    | 40.5     | 5000          | 0              | 0             |
| 34  | 90          | 0                       | Z negative         | 160   | 96.2   | 160      | -96.2    | -10      | 5000          | 0              | 30            |
| 35  | 96          | 0                       | Z negative         | 143.5 | -222.4 | 143.5    | 222.4    | 35.167   | 5000          | 0              | 30            |
| 36  | 151         | 8                       | Z negative         | 0     | 0      | 0        | 0        | 0        | 5000          | 0              | 60            |
| 37  | 152         | 0                       | Z negative         | 0     | -32    | 0        | 32       | 0        | 5000          | 0              | 60            |
| 38  | 153         | 0                       | Z negative         | 0     | -64    | 0        | 64       | 0        | 5000          | 0              | 60            |
| 39  | 154         | 0                       | Z negative         | 0     | -96    | 0        | 96       | 0        | 5000          | 0              | 60            |
| 40  | 155         | 0                       | Z negative         | 0     | -128   | 0        | 128      | 0        | 5000          | 0              | 30            |
| 41  | 156         | 0                       | Z negative         | 0     | -160   | 0        | 160      | 0        | 5000          | 0              | 30            |
| 42  | 157         | 0                       | Z negative         | 0     | -192   | 0        | 192      | 0        | 5000          | 0.2            | 30            |
| ..  | ..          | -                       | -                  | ..    | ..     | ..       | ..       | -        | ....          | ..             | ..            |

## Accuracy adjustment of Z-direction

To adjust the depth of the notch, adjust the Z offset of the spindle;

[+] add when it's deeper and [-] subtract if it's shallower;





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